Statistical Inference & Hypothesis Testing Report

Analyst: Portfolio Toolkit | **Date:** June 02, 2026 | **Framework:** AP Statistics Standard

1. State

Define the hypotheses and significance level to evaluate the claims of the A/B test.

Significance Level (α): 0.05

Hypotheses (Symbols):

$$H_0 : \text{Variables are independent} \quad H_a : \text{Variables are dependent}$$

Null Hypothesis (H_0): The categorical variables are independent; the distribution of responses is similar across groups.
Alternative Hypothesis (H_a): The categorical variables are dependent; the distribution of responses is significantly different across groups.

2. Plan

Identify the appropriate statistical procedure and verify its mathematical assumptions.

Recommended Procedure: Chi-Square Test of Independence

Assumption / Condition Checked	Status & Diagnostic Details
Randomness	Categorical frequencies collected from random sampling.
Independence	Observations represent mutually exclusive categorical responses.
Large Sample (Expected counts)	All expected cell counts ≥ 5? **Yes** (Minimum expected cell count = 22.86).

■ **CONDITIONS MET**

All mathematical conditions for the Chi-Square Test of Independence are satisfied. Parametric results are highly reliable.

3. Do

Calculate sample metrics, standard error, test statistic, and the resulting p-value.

Sample Descriptive Statistics:

Cell Index (Row, Col)	Observed Count (O)	Expected Count (E)	Residual Contribution (O-E)²
Row 1, Col 1	60	55.24	0.4105
Row 1, Col 2	115	121.90	0.3911
Row 1, Col 3	25	22.86	0.2009
Row 2, Col 1	56	60.76	0.3732
Row 2, Col 2	141	134.10	0.3555
Row 2, Col 3	23	25.14	0.1826

Calculations & Mathematical Formulations:

• *Chi2 Stat*:

$$\chi^2 = \sum \frac{(O-E)^2}{E} = 1.9138$$

• *Df*:

$$df = 2$$

Statistical Parameter	Value
Test Statistic (χ^2)	1.91385
p-value	0.38407
Degrees of Freedom (df)	2

4. Conclude

State the final decision and offer a business interpretation based on the statistical results.

Compare p-value to α: 0.38407 >= 0.05. Since the p-value is greater than or equal to the significance level, we **fail to reject the null hypothesis (H_0)**. There is insufficient evidence to conclude there is a significant difference between the variations.

■ **DECISION: FAIL TO REJECT NULL HYPOTHESIS**

Business Action Recommendation: No significant association was detected. User distributions are statistically homogeneous across groups. Standardize the layout/offering.